



RAMCO INSTITUTE OF TECHNOLOGY

Department of Civil Engineering



STRUCTA 2024-25

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PRESENTATION

Vol
5

MAGAZINE 2024-25

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ABOUT THE DEPARTMENT

- The Department of Civil Engineering was established in the year 2013. At present, it offers undergraduate programme, B.E. in Civil Engineering with a sanctioned intake of 60 students. The Department broadly covers important fields such as Surveying, Geotechnical Engineering, Structural Engineering, Hydraulics and Water Resources Engineering, Construction Materials and Practices, Environmental Engineering and Transportation Engineering.
- The Department has well established modern laboratories to cater to the needs of academic and research activities. It is well equipped with computational facilities and resources both in terms of hardware and software. The Department constantly facilitates students from other departments of the Institute as a gesture of support to their academic, co-curricular and research desires.
- The Department has well-qualified and committed faculty members, lab technicians and modern infrastructure in the form of elegant and smart classrooms, spacious laboratories, latest machinery and equipment. Students are encouraged to participate in co-curricular and extra-curricular activities to acquire leadership qualities. The Civil Engineering Association RIT STRUCTA has been functioning for the benefit of the students. It serves as a platform for students to make technical presentations, gain knowledge about recent advancements and bring together novel ideas. This Association aims at bridging the gap between the academic and the industry by inviting renowned industrialists and academicians to interact with students. Guest lectures, industrial visits, site visits, etc., are organized and value added courses are conducted on regular basis to ensure that students are conversant with recent developments in the field of Civil Engineering.
- The Department encompasses professional societies such as Indian Concrete Institute (ICI), ISTE chapters, which provides students a platform to acquire knowledge in various technical activities. It offers consultancy services in the fields of Structural Engineering, Geotechnical Engineering, Environmental Engineering, Surveying and Highway Engineering.

VISION

- Create knowledgeable and professionally competent Civil Engineers who are capable of providing global solutions to the problems related to Civil Engineering.

MISSION

- Impart high quality education and technical skills to transform students into Professional Civil Engineers.
- Establish state-of-the-art infrastructure for updating knowledge in recent developments and cutting edge technology.
- Bridge the gap between academics and industry by practical training and providing solutions for design projects related to Civil Engineering.
- Motivate the faculty members and students in the teaching learning process for achieving excellence to meet the global standards.
- Produce graduates with high professional and ethical standards for fulfilling the needs of the industry and society.

PROGRAMME EDUCATIONAL OBJECTIVES

- Graduates of the programme will be professionally competent and apply the fundamental concepts of mathematics, science and Civil Engineering to solve problems in Civil Engineering.
- Graduates of the programme will perform their responsibilities as professional Civil Engineers with quality and ethics Graduates of the programme, as part of an organization or as Entrepreneurs, will continue to learn with evolving technologies.
- Graduates of the programme will innovate and provide global solutions for a sustainable built environment.
- Graduates of the programme will contribute to the development of sustainable Infrastructure for the betterment of society.

PROGRAMME SPECIFIC OUTCOMES

- Communicate and present civil engineering projects effectively.
- Use the techniques, skills, and modern engineering tools necessary for civil engineering practice and project management.
- Provide sustainable solutions to civil engineering problems.
- Perform as design consultants in construction industry for the design of civil engineering structures.

PROGRAMME OUTCOMES

- Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

CONTENT

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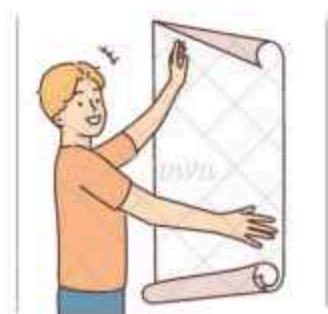
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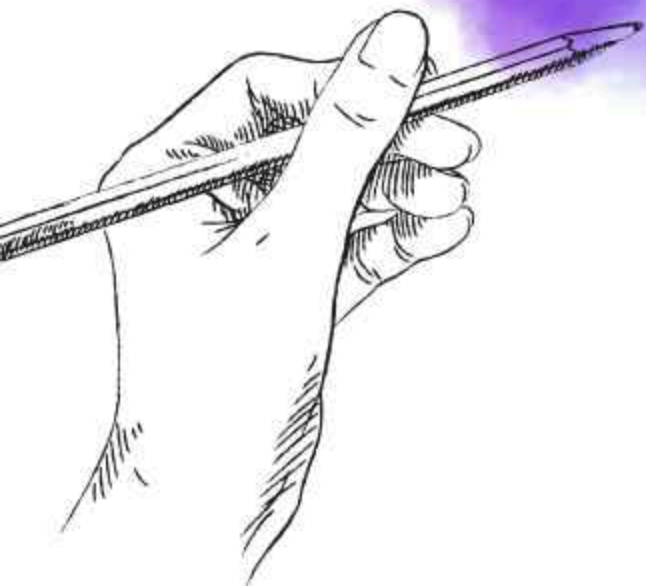
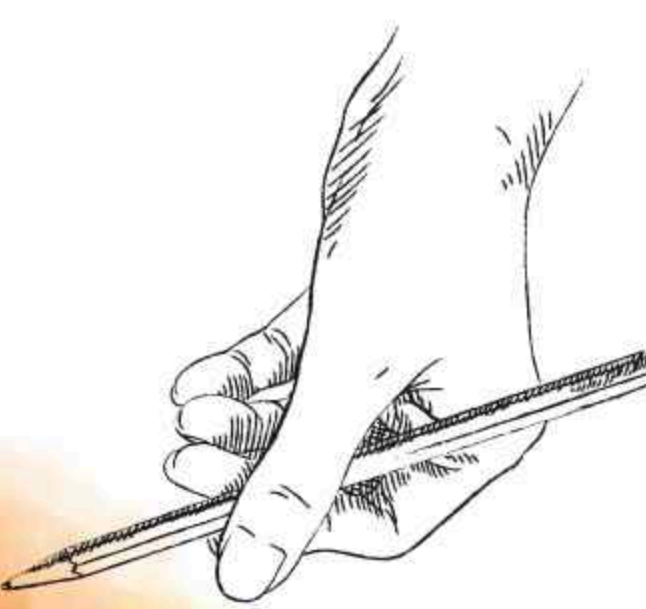


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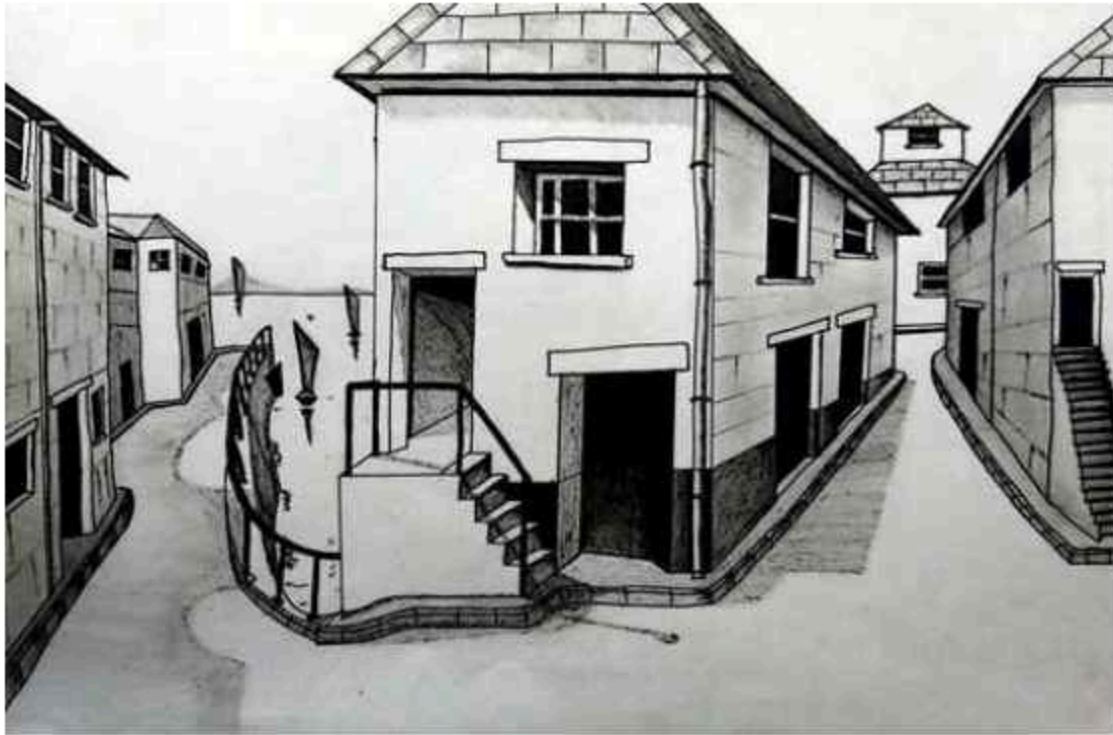


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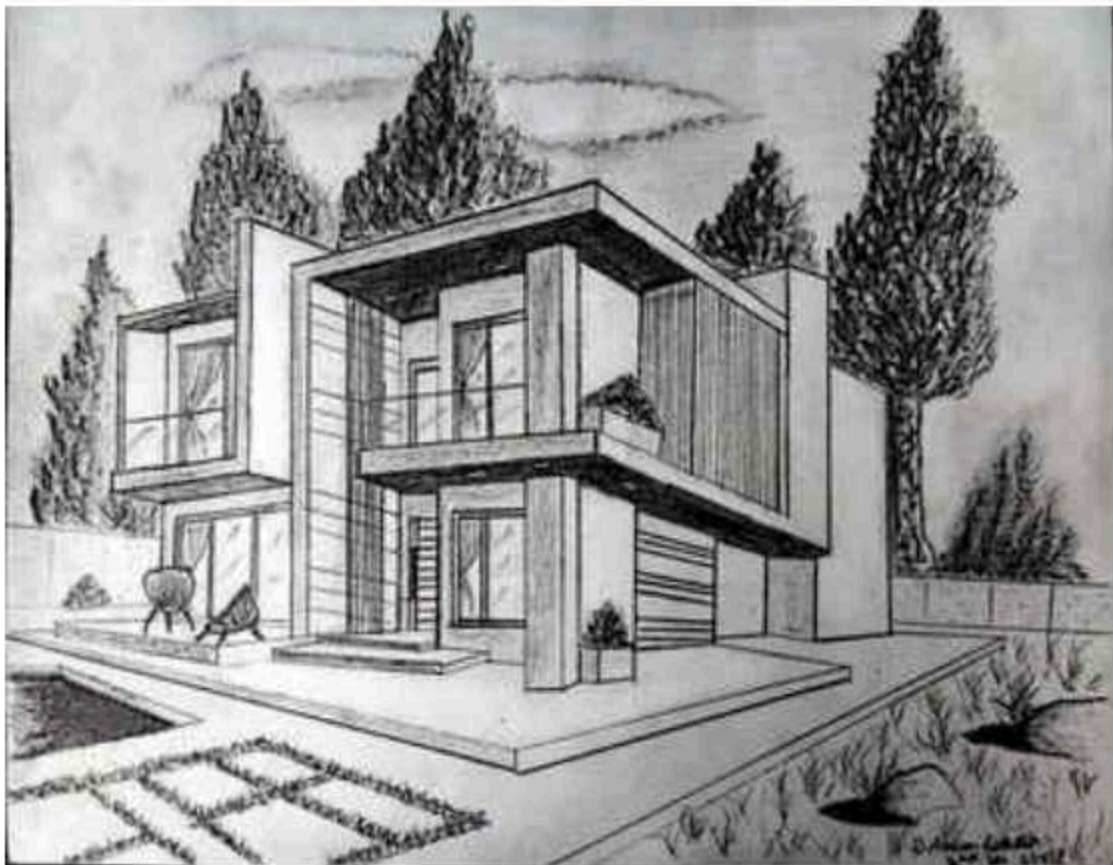




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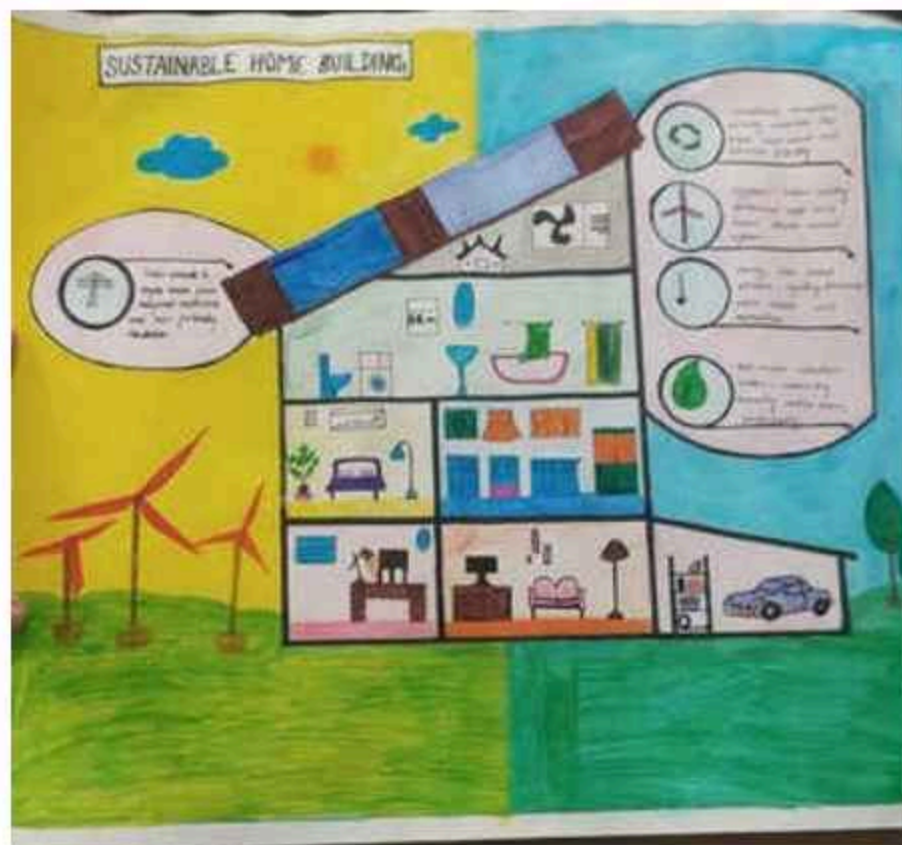
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R Arun Lokesh / III YEAR



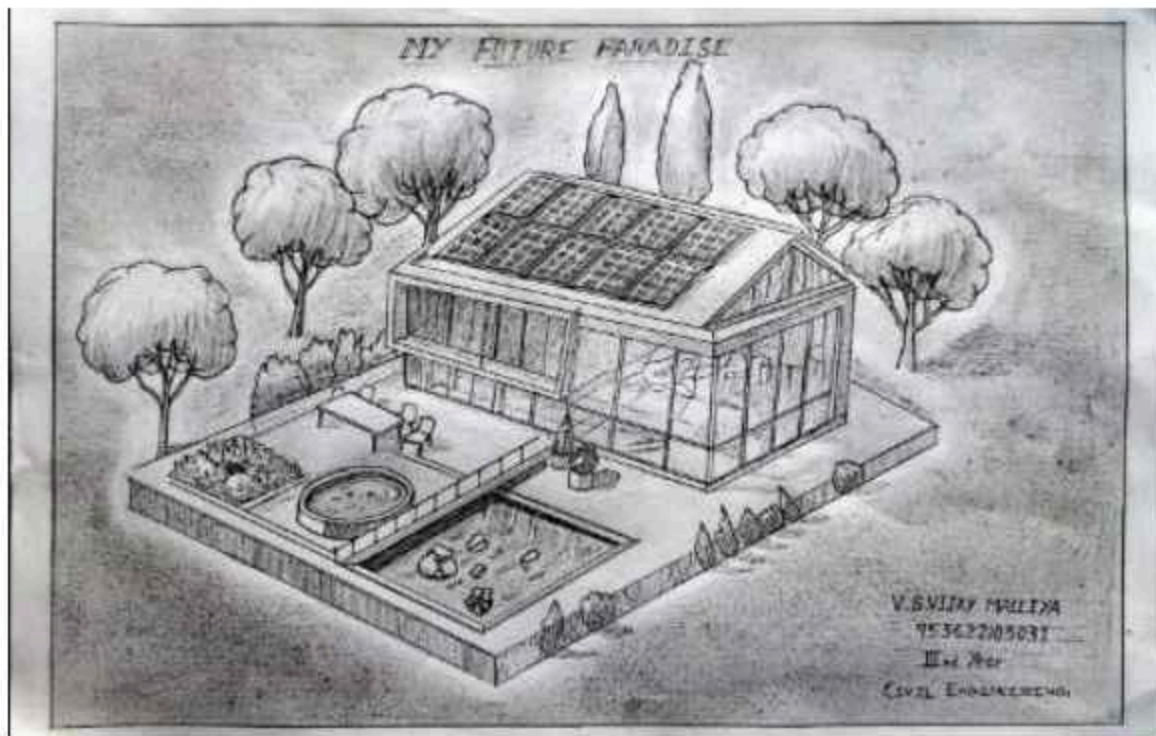
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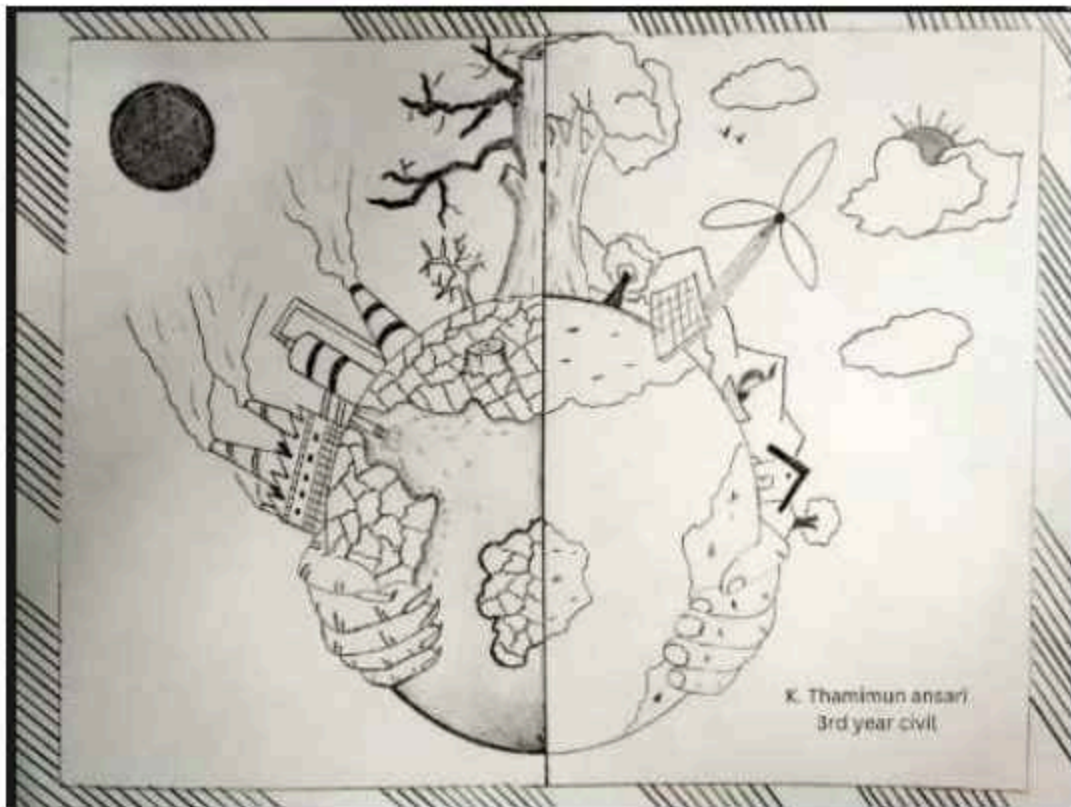
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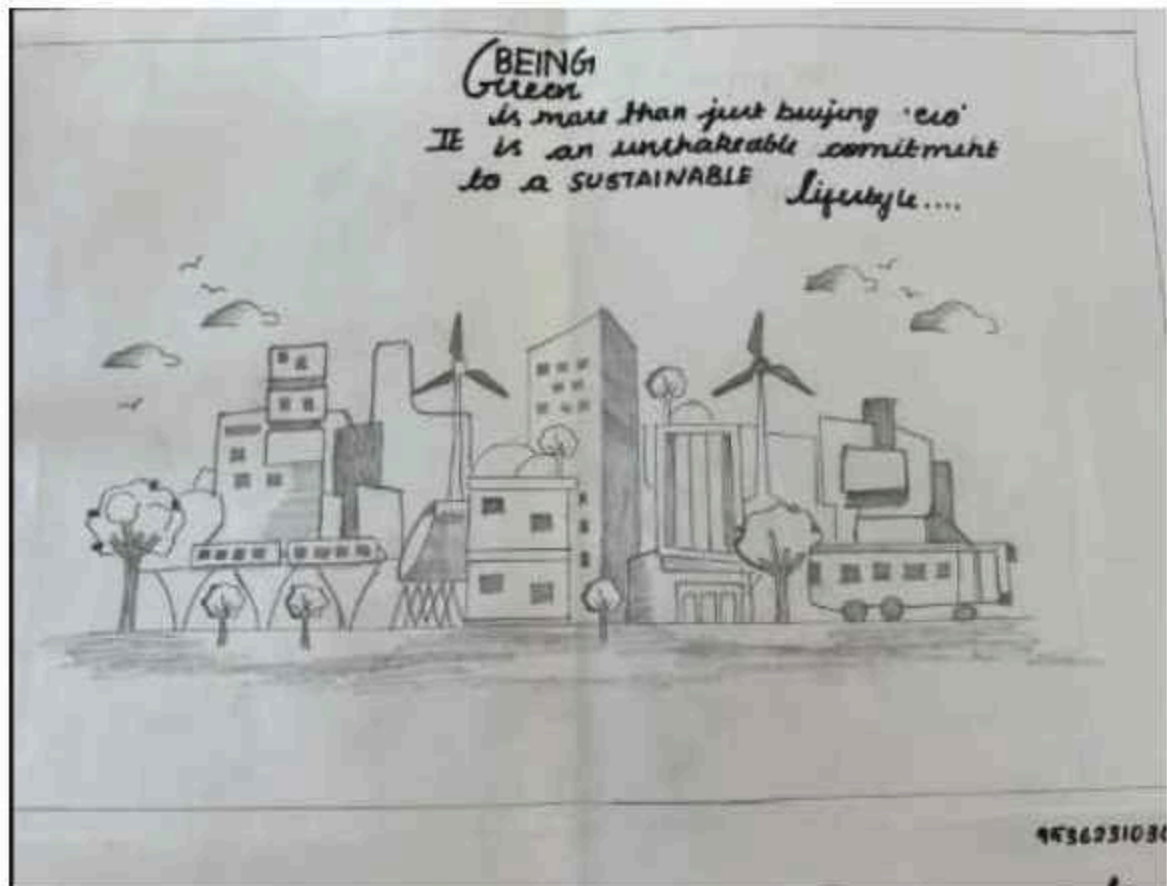
VIJAY MALLIYA V S/ IIIYEAR



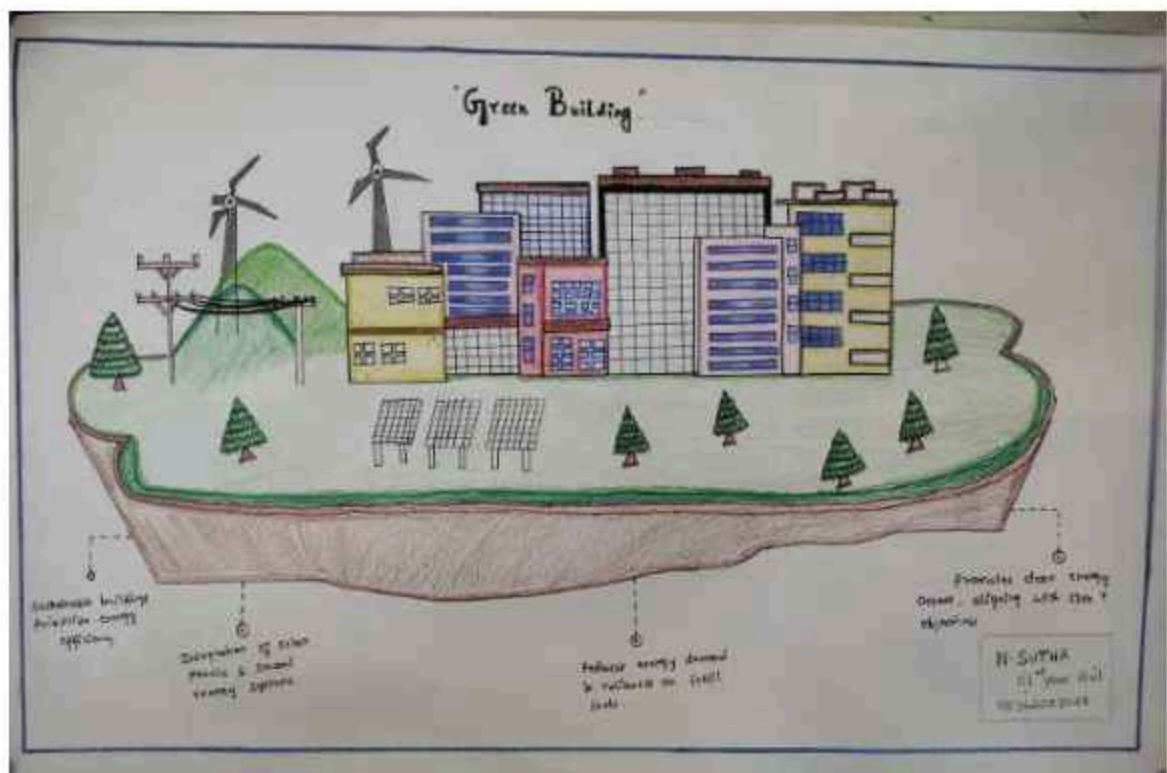
THAMIMUN ANSARI K
III YEAR



RAMYA SUDHA N / III YEAR



HIMAJAA J / II YEAR



SUTHA N /III YEAR



ALAGU MURUGAN / IV YEAR



Udhaya Ganesh T/ I YEAR



Sudharshan R / I YEAR

A woman with blonde hair, wearing a light-colored jacket, is holding a large white rectangular sign with a black border. The sign is positioned in front of her, and she is looking at the camera with a slight smile. The background is a green wall with a window. The word "POSTERS" is written in bold, black, italicized capital letters on the sign.

POSTERS

AI IN CONCRETE TECHNOLOGY

AI is revolutionizing concrete technology by improving material design, quality control, predictive maintenance, and construction efficiency. Here's how AI is impacting different aspects of concrete technology:

AI IN MIX DESIGN OPTIMIZATION

- AI systems analyze large datasets to optimize concrete mix proportions (cement, aggregates, water) tailored for strength, durability, and sustainability.
- Machine learning helps reduce cement content while maintaining performance, reducing CO₂ emissions.



1



QUALITY CONTROL & DEFECT DETECTION

- AI-powered image recognition detects cracks, voids, and other defects in concrete structures using drone and camera footage.
- Machine learning predicts early-age concrete strength based on curing conditions, avoiding tedious test testing.

PREDICTIVE MAINTENANCE & STRUCTURAL HEALTH MONITORING

- AI processes real-time data from IoT sensors embedded in concrete to monitor stress, temperature, and strain.
- AI models predict structural health, identifying risks, enabling proactive maintenance and preventing structural failure.



3



SUSTAINABILITY & CO₂ REDUCTION

- AI optimizes concrete formulations by suggesting alternative binders (fly ash, slag, or geopolymers).
- AI-powered simulations assess the long-term environmental impact of different concrete formulations.

CONSTRUCTION SITE SAFETY

- AI-powered vision systems monitor construction sites for unsafe practices, identifying potential accidents or equipment malfunctions.
- AI helps in real-time monitoring of workers, ensuring safety protocols are followed.



5

AI-POWERED BIM: REVOLUTIONIZING CONSTRUCTION PLANNING



WHAT IS IT?

AI-powered BIM (Building Information Modeling) is an advanced technology that integrates AI with BIM to automate design, enhance collaboration, optimize costs, monitor construction, and improve sustainability in the building industry.

1. DESIGN AUTOMATION

AI automates 3D model generation, reduces errors, and optimizes designs.



2. REAL-TIME COLLABORATION

AI-powered BIM enables cloud collaboration, clash detection, and document management.



3. COST OPTIMIZATION

AI predicts costs, reduces material wastage, and suggests budget-friendly solutions.

4. SMART CONSTRUCTION MONITORING

AI integrates with IoT and drones for site monitoring, safety analysis, and progress tracking.



5. SUSTAINABILITY & EFFICIENCY

AI optimizes energy use, selects eco-friendly materials, and supports green building strategies.



PPE DETECTION IN CONSTRUCTION SITE



USING AI

WHAT IS PPE DETECTION?

Using personal protective equipment (PPE Detection) has always been a critical part of the work process in manufacturing, construction, and healthcare facilities. Workers in these environments meet hazards every day, such as chemicals, fall from height and so on.

PRECAUTION RELEASE RISK



HOW IT'S WORK WITH AI?

- Detects PPE in images or videos: AI software analyzes visual data to identify PPE.
- Automatic and real-time detection: PPE detection happens instantly, without manual intervention.
- Sends notifications for non-compliance: Alerts safety managers if a worker is not wearing required PPE.

OSHA STANDARDS:

AI-Enhanced Occupational Health Compliance Safety Audits. By using IoT sensors, AI can track and audit every individual worker on multiple levels, ensuring that workplaces adhere to safety standards, minimize legal risks, and promote a culture of compliance.



BENEFITS:



- Reduce PPE violations and injuries
- Automates PPE inspection
- Improves worker safety
- Increases productivity

Team Members:

N. Sotha
S. Nishanthi
M. Asmath
Marzia



MINIATURE TETRAPOD MOULD (3D PRINTED MODEL)

Introduction

Tetrapods are four-legged concrete structures widely used in coastal engineering to prevent coastal erosion and dissipate wave energy. Their interlocking design resists displacement even under strong wave action.



Objective

To design and fabricate a miniature tetrapod mould using 3D printing technology for demonstration and academic purposes



Process

- 3D CAD modeling of tetrapod structure.
- Conversion of model into printable STL format.
- Fabrication using 3D printer (PLA/ABS material).
- Finished product: miniature mould prototype suitable for academic and lab-scale casting.



Applications

- Coastal protection studies.
- Hydraulic and wave tank experiments.
- Demonstrations in civil engineering laboratories.
- Educational and research purposes

BY
VIJAYA SIVA SANKAR



ARTIFICIAL INTELLIGENCE IN CIVIL ENGINEERING



Enhancing AI today means building a stronger, smarter, and more efficient tomorrow.

ML for Structure health monitoring

Machine Learning (ML) is increasingly used for structure health monitoring (SHM) to detect anomalies, predict damage or deterioration in structures like bridges, buildings, and dams. It helps automate inspection, improve accuracy, and reduce costs over time.



Prevent cost overruns

AI-based Cost Estimation & Control

- AI-driven predictive analysis for accurate cost estimation
- Machine learning models for optimizing resource allocation



Automated Risk Reduction

AI for Risk Prediction and Mitigation

- AI models analyzing historical data to predict risks
- Automated risk monitoring with computer vision and sensors

Optimizations in Civil Engineering

- AI-powered safety compliance monitoring
- AI-driven decision support systems for project managers



Increase Productivity on job sites



AI-powered Solutions for Job Site Efficiency

- Robotics & automation in construction processes
- AI-based scheduling and resource optimization

Smart Construction Technologies

- AI-driven equipment tracking and fleet management
- Drones for real-time site monitoring and progress tracking

CONCLUSION

Artificial Intelligence is revolutionizing the civil engineering industry by enhancing efficiency, reducing risk, and optimizing resource.

- From structural health monitoring to cost management, risk reduction, and job site productivity, AI-driven solutions are making construction smarter and safer.

By leveraging machine learning, predictive analytics, and automation, civil engineers can build more resilient, cost-effective, and sustainable infrastructure.

- As AI technology continues to evolve, its integration with civil engineering will shape the future of smart construction, improving safety, quality, and project outcomes.



Shaping the Future: AI and ML in Civil Engineering Solutions

Presented by
Ananya Mishra II
Aravind Kumar II
Ananya Mishra II



ARTIFICIAL INTELLIGENCE IN SMART CONSTRUCTION



AUTOMATED DESIGN & PLANNING

- AI-powered BIM (Building Information Modeling) optimizes project design.
- Generative design helps architects create efficient structures.

PREDICTIVE MAINTENANCE & RISK ASSESSMENT

- AI analyzes historical data to predict equipment failures.
- Helps in proactive maintenance, reducing downtime and costs.



CONSTRUCTION AUTOMATION & ROBOTICS

- AI-driven robots assist in repetitive tasks like bricklaying.
- Autonomous machinery reduces human error and increases productivity.

PROJECT MANAGEMENT & COST OPTIMIZATION

- AI forecasts delays and budget overruns.
- Enhances decision-making by analyzing past project data.



SUSTAINABLE CONSTRUCTION WITH AI

- AI optimizes material usage, reducing waste.
- Smart energy management systems improve building efficiency.

CONCLUSION

- AI is transforming construction by making it smarter, safer, and more efficient.
- Future advancements will lead to fully autonomous and sustainable construction processes.



ARTIFICIAL INTELLIGENCE IN CIVIL ENGINEERING



INTRODUCTION

Artificial intelligence (AI) is used in civil engineering to improve construction efficiency, safety, and design. AI can also help with cost estimates and risk assessment.



AI TOOLS USED IN CIVIL ENGINEERING

ClickUp: Helps with project management and productivity
Fieldwire: A construction management platform that helps with task management, document sharing, and reporting
Procore: A construction management tool



TECHNIQUES

In civil engineering, AI techniques like machine learning, computer vision, natural language processing, evolutionary computation, and deep learning are used for applications such as construction management, structural health monitoring, geotechnical engineering,



USES OF AI

Generative AI can also be used for real-time monitoring and inspection of construction sites. For example, an AI-powered system that uses cameras and sensors to monitor the progress of a construction project, and compare it to the original plans and blueprints.



CONCLUSION

Artificial Intelligence has been effectively applied to a variety of civil engineering applications, including prediction and risk management. Artificial Intelligence used in Civil engineering is important in the construction, maintenance, and management of several components of civil infrastructure.



>>>>>>>>> ARTIFICIAL INTELLIGENCE IN CIVIL ENGINEERING: TRANSFORMING THE FUTURE

WHY AI IN CIVIL ENGINEERING?

AI improves efficiency, accuracy, and decision-making in planning, design, construction, and maintenance of infrastructure.



APPLICATIONS OF AI IN CIVIL ENGINEERING

- Smart Construction & Robotics
- AI in Design & Planning
- Traffic & Transportation Management



BENEFITS OF AI IN CIVIL ENGINEERING

- Increased Efficiency: Automates repetitive tasks, reducing human effort.
- Cost Reduction: Minimizes material waste and project delays.
- Improved Safety: AI-powered monitoring reduces workplace hazards.
- Sustainability: Optimizes resource use and reduces environmental impact.

CONCLUSION

- AI is revolutionizing civil engineering, making it smarter, safer, and more sustainable.
- Engineers must embrace AI technologies to stay ahead in innovation.

PRESENTED BY
U.S. VIJAY MALLIYA
M. SRIKRISHNA
G. SETHUAYANAR
S.R. SUDHARSAN

MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE USE IN CIVIL ENGINEERING

Embracing AI today means building a stronger, smarter, and more efficient tomorrow

ML for Structure health monitoring

Role of Machine Learning in SHM:

- Data-driven analysis from IoT sensors (vibration, strain, temperature)
- AI-powered predictive maintenance to prevent failures

Case Studies & Real-world Applications:

- AI in bridge and building monitoring
- Early detection of structural defects using deep learning



PREVENT COST OVER RUNS

Challenges in Cost Management:

- Budget overruns due to design changes, material costs, and labor issues
- Impact of poor forecasting and planning

AI-based Cost Estimation & Control:

- AI-driven predictive analytics for accurate cost estimation
- Machine learning models for optimizing resource allocation

AUTOMATED RISK REDUCTION

AI for Risk Prediction and Mitigation:

- AI models analyzing historical data to predict risks
- Automated site monitoring with computer vision and drones

Implementation in Civil Engineering:

- AI-powered safety compliance monitoring
- AI-driven decision support systems for project managers



INCREASE PRODUCTIVITY ON JOB SITES

AI-powered Solutions for Job Site Efficiency:

- Robotics & automation in construction processes
- AI-based scheduling and resource optimization

Smart Construction Technologies:

- AI-driven equipment tracking and fleet management
- Drones for real-time site monitoring and progress tracking

CONCLUSION

Artificial Intelligence is revolutionizing the civil engineering industry by enhancing efficiency, reducing risks, and optimizing resources. From structural health monitoring to cost management, risk reduction, and job site productivity, AI-driven solutions are making construction smarter and safer.

By leveraging machine learning, predictive analytics, and automation, civil engineers can build more resilient, cost-effective, and sustainable infrastructure. As AI technology continues to evolve, its integration into civil engineering will shape the future of smart construction, improving safety, quality, and project outcomes.



Shaping the Future: AI and ML in Civil Engineering Solutions

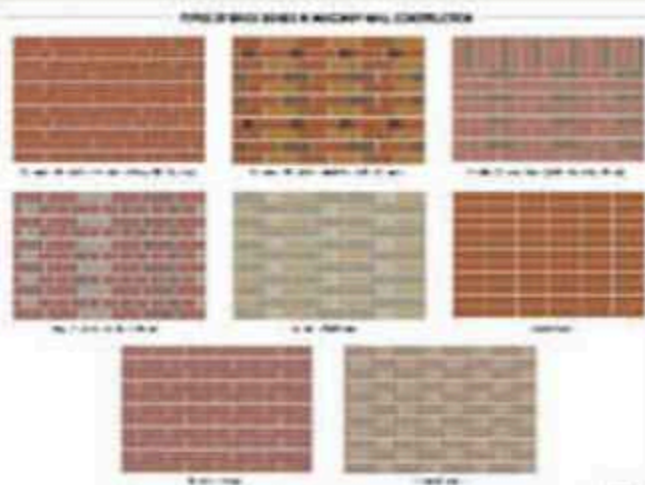
BRICK MASONRY

DEFINITION

It is build by placing bricks in mortar in a systematic manner to construct solid mass that withstand exerted loads



TYPES OF BRICK MASONRY



MATERIALS USED

- 1.Mortar Mix or Manson mix bricks
- 2.tape measure
- 3.hammer
- 4.Hose, level or theodolite
- 5.Trowel level
- 6.Wheelbarrow
- 7.Goggles
- 8.Jointer

MANUFACTURING OF BRICKS

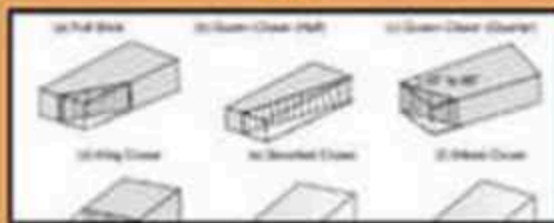
To manufacture a brick masonry wall, engineers would like raw materials like fly ash ,gypsum ,cement and sand that area unit manually fed into a pan mixture wherever water is supplementary within the needed proportion for intimate commixture

ADVANTAGES

- 1.Brick are resistant to burning
- 2.safer option for construction
3. High resistant to natural disaster

WEATHER LIMITATIONS

- Heavy rain and freezing temperature should be avoided
- Freezing temperatures can causes cracking and damage
- Delay may occur due to weathers limitations



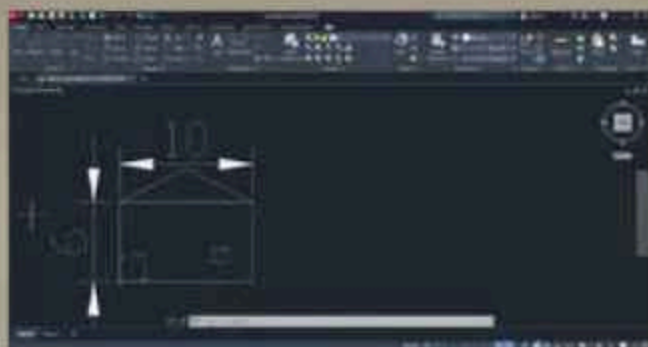


AI IN CIVIL DRAWINGS



AI prompts are transforming civil engineering by making drawing creation faster and more accurate. By simply describing a structure, engineers can generate detailed CAD drawings instantly, saving time on manual drafting.

This technology helps speeds up the developing process of drawings while reduce errors by ensuring precise measurements and standardized design elements, improving overall project quality and communication.



THIS IS AN SIMPLE EXAMPLE OF CREATING AN AI APPLICATION FOR DEVELOPING CAD DRAWINGS CALLED CADAI.

BY
ATHEEQUR RAHMAN

3D ELEVATION





KAVI KUMAR / IV YEAR



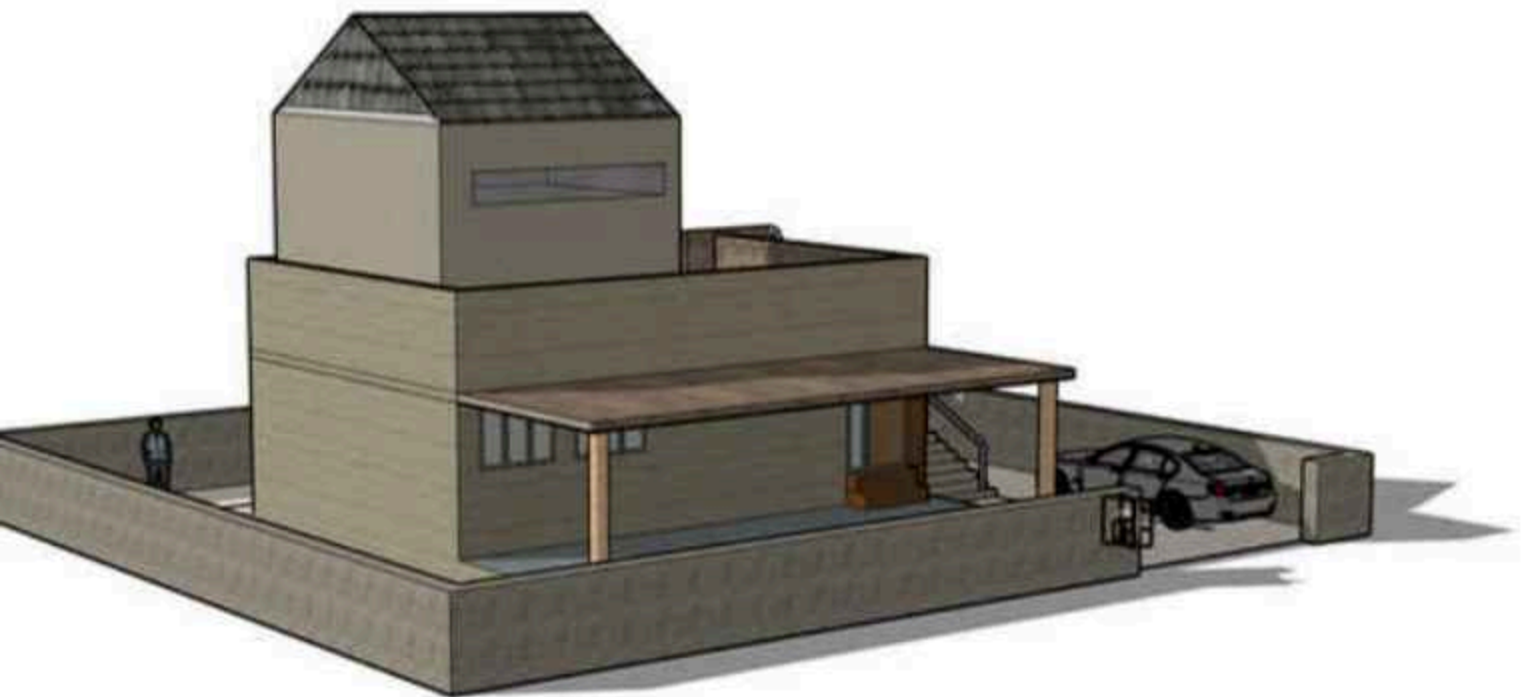
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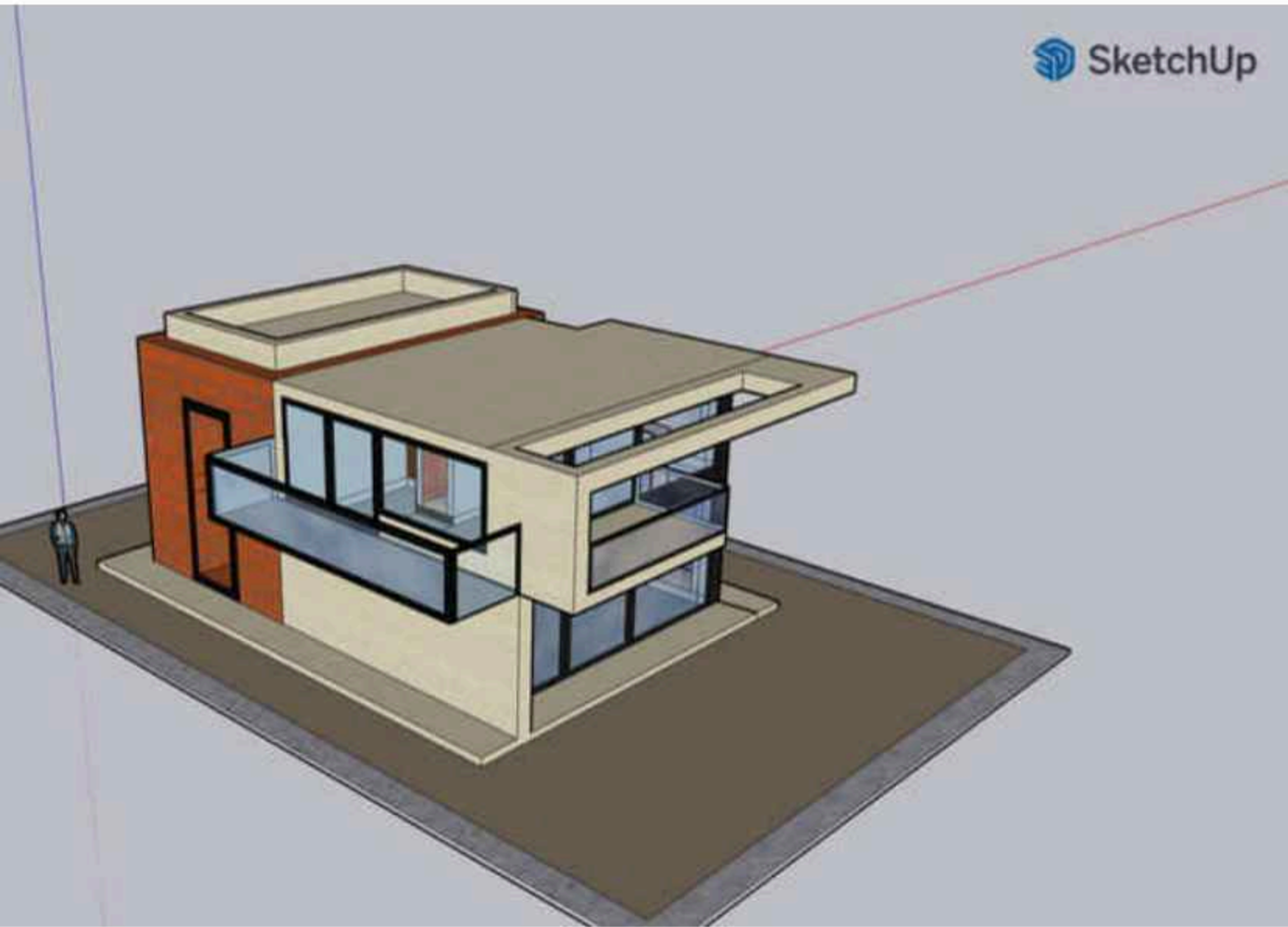
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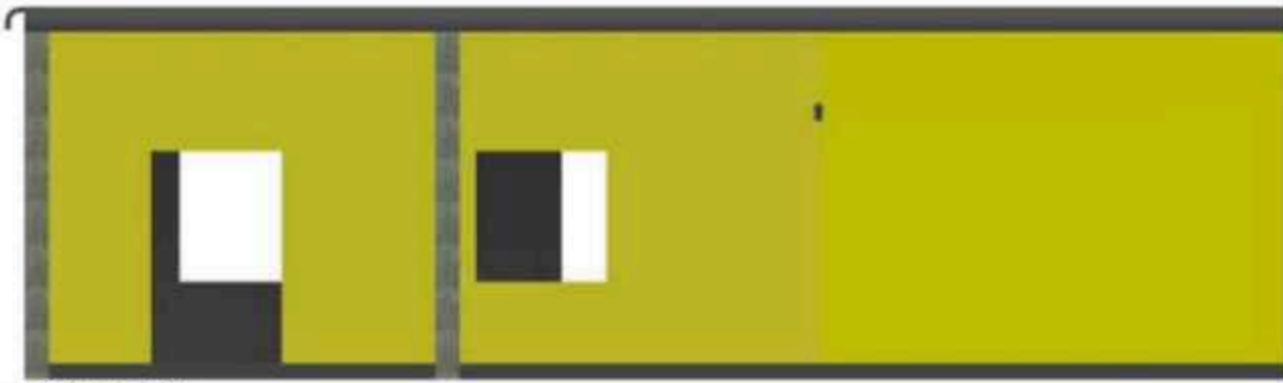
PANDI GANESH / II YEAR



VIJAYA SIVA SANKAR / I YEAR



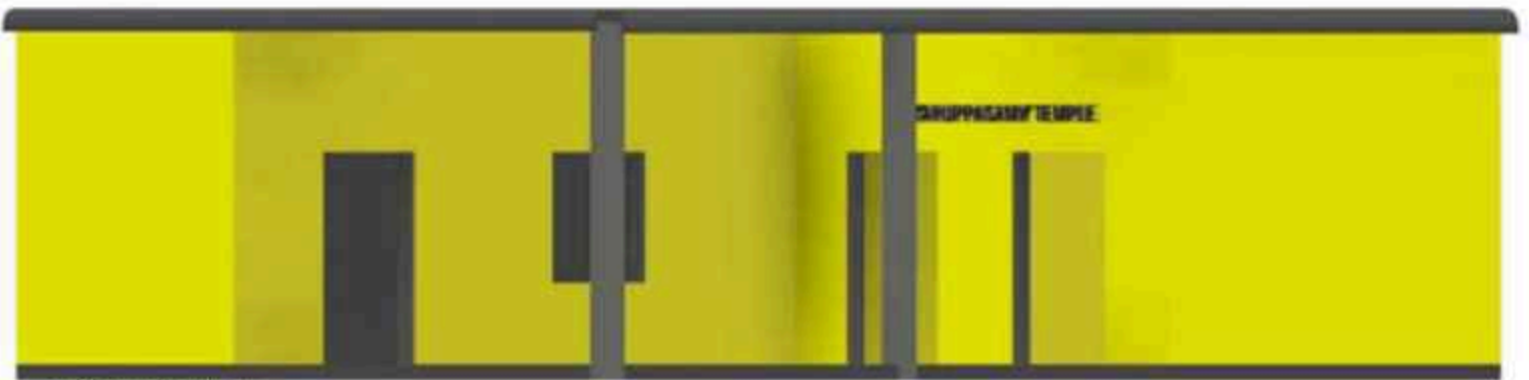
VIJAYA SIVA SANKAR



① SOUTH VIEW

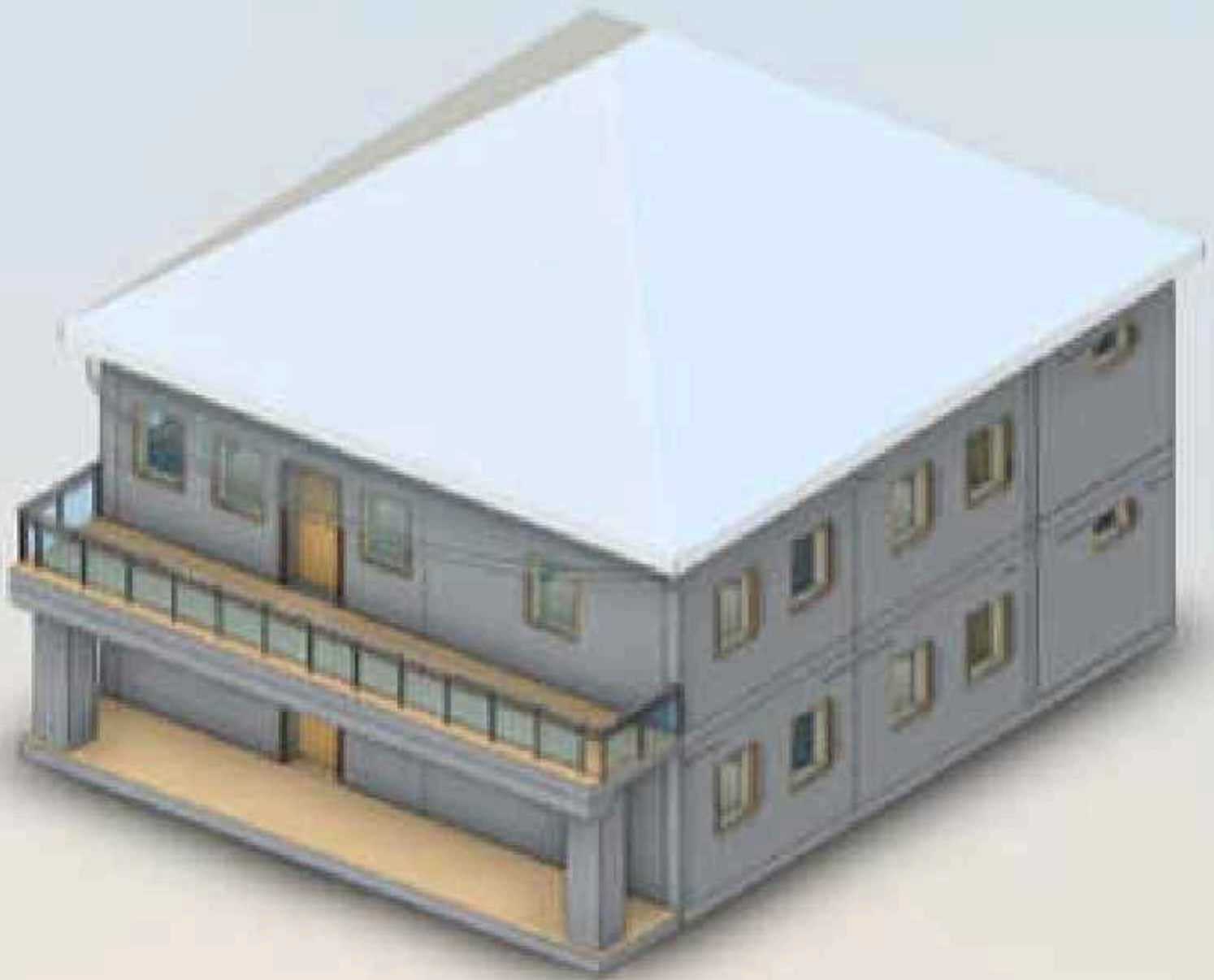


② WEST VIEW

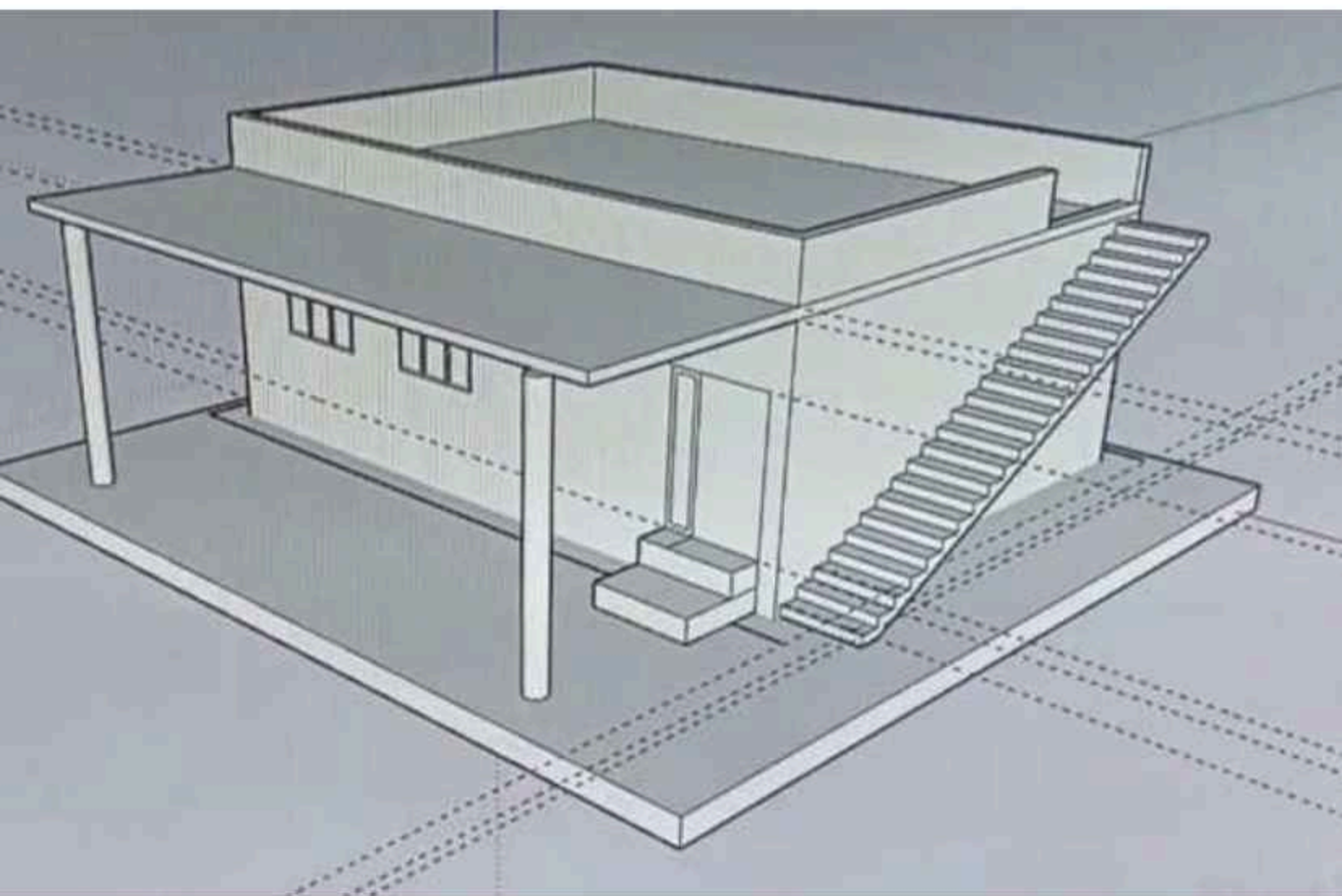


③ SOUTH WEST VIEW

VEL PRASATH / II YEAR

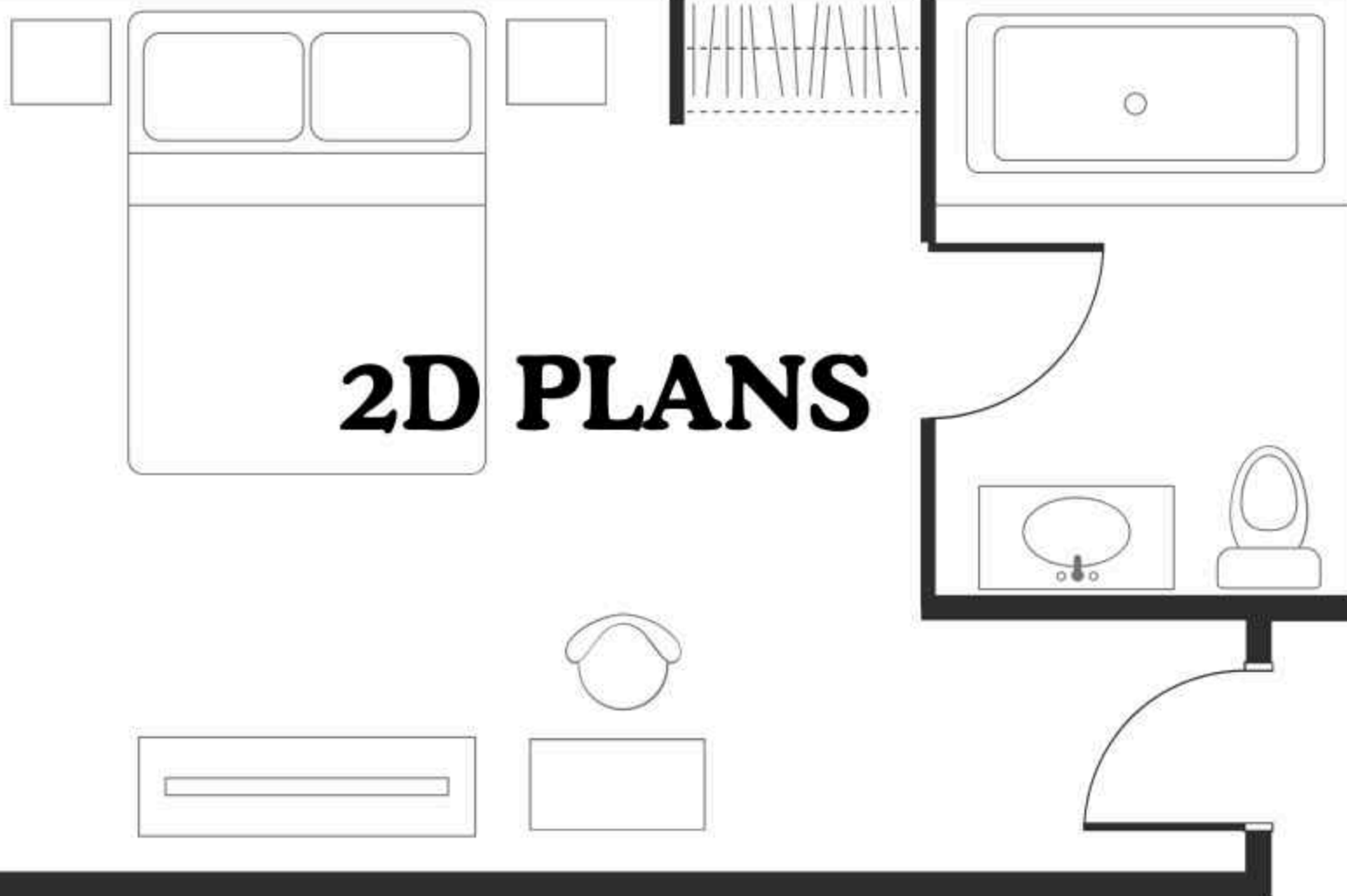


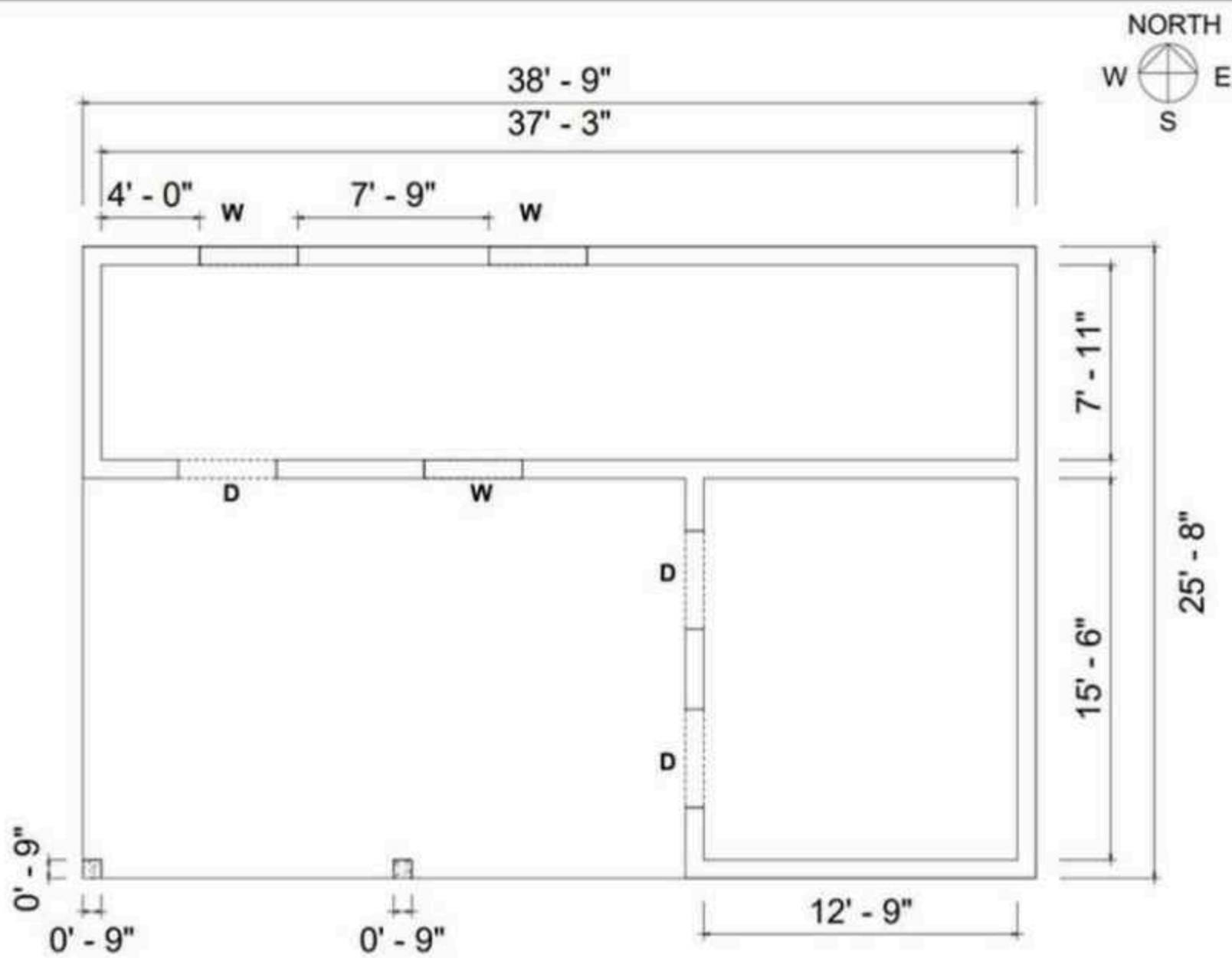
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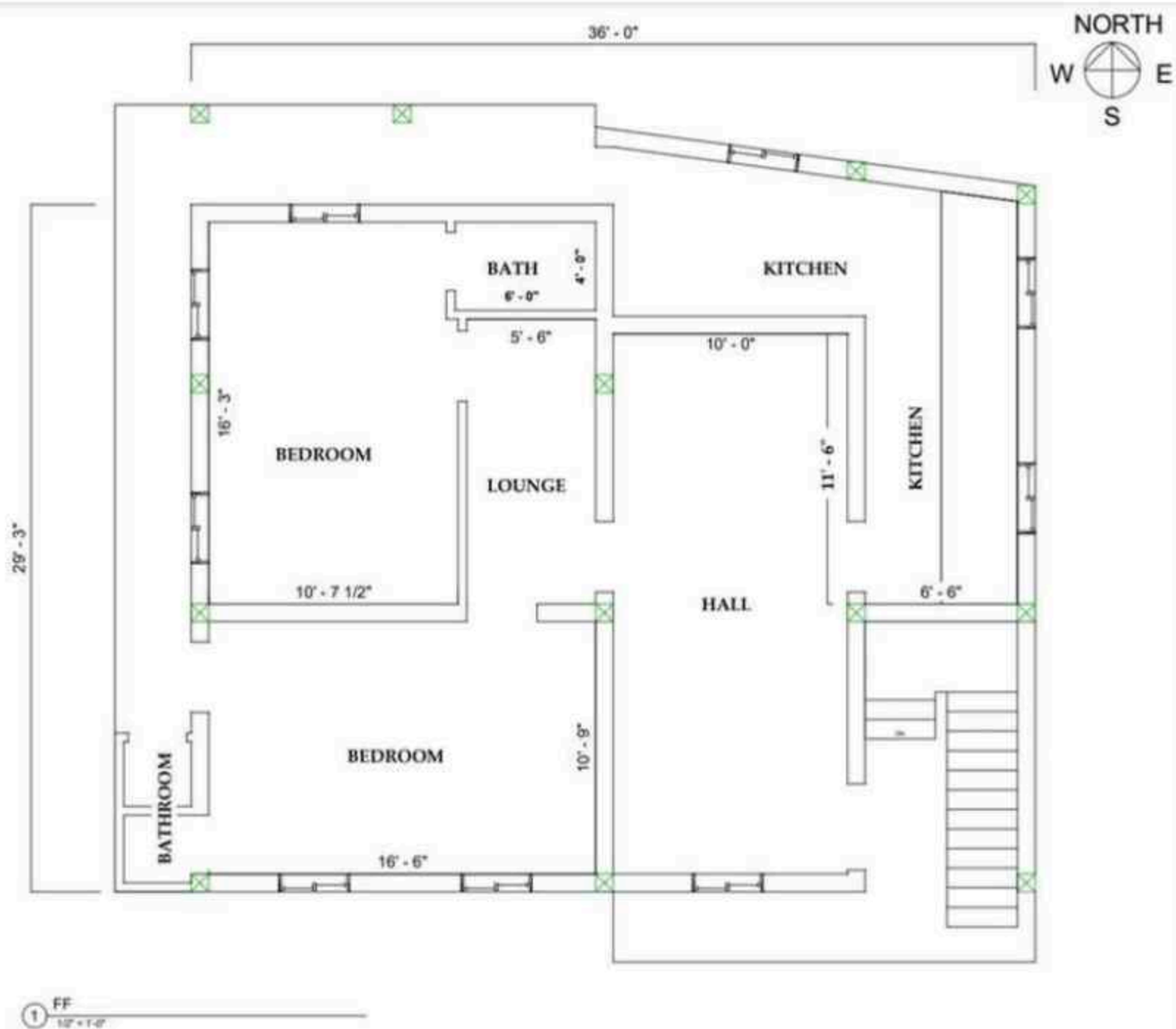
R.DHANUSRI / I YEAR

2D PLANS

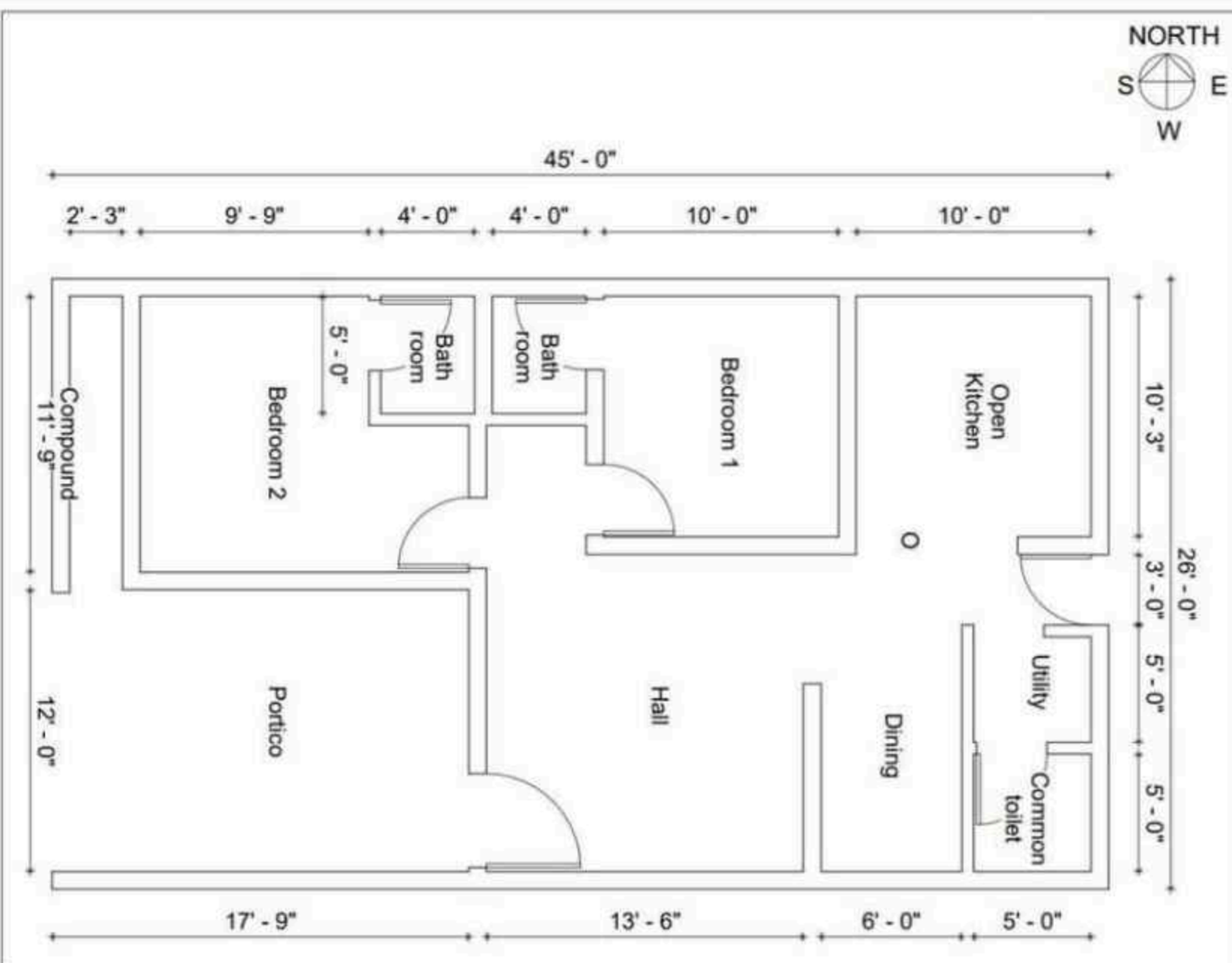




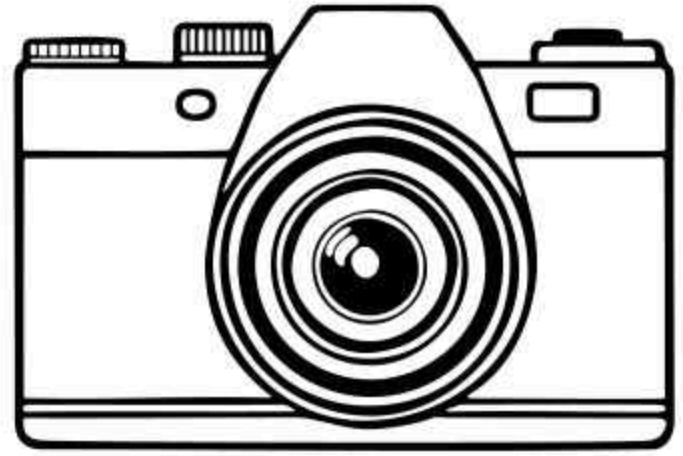
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HARI AARA MOORTHY / II YEAR



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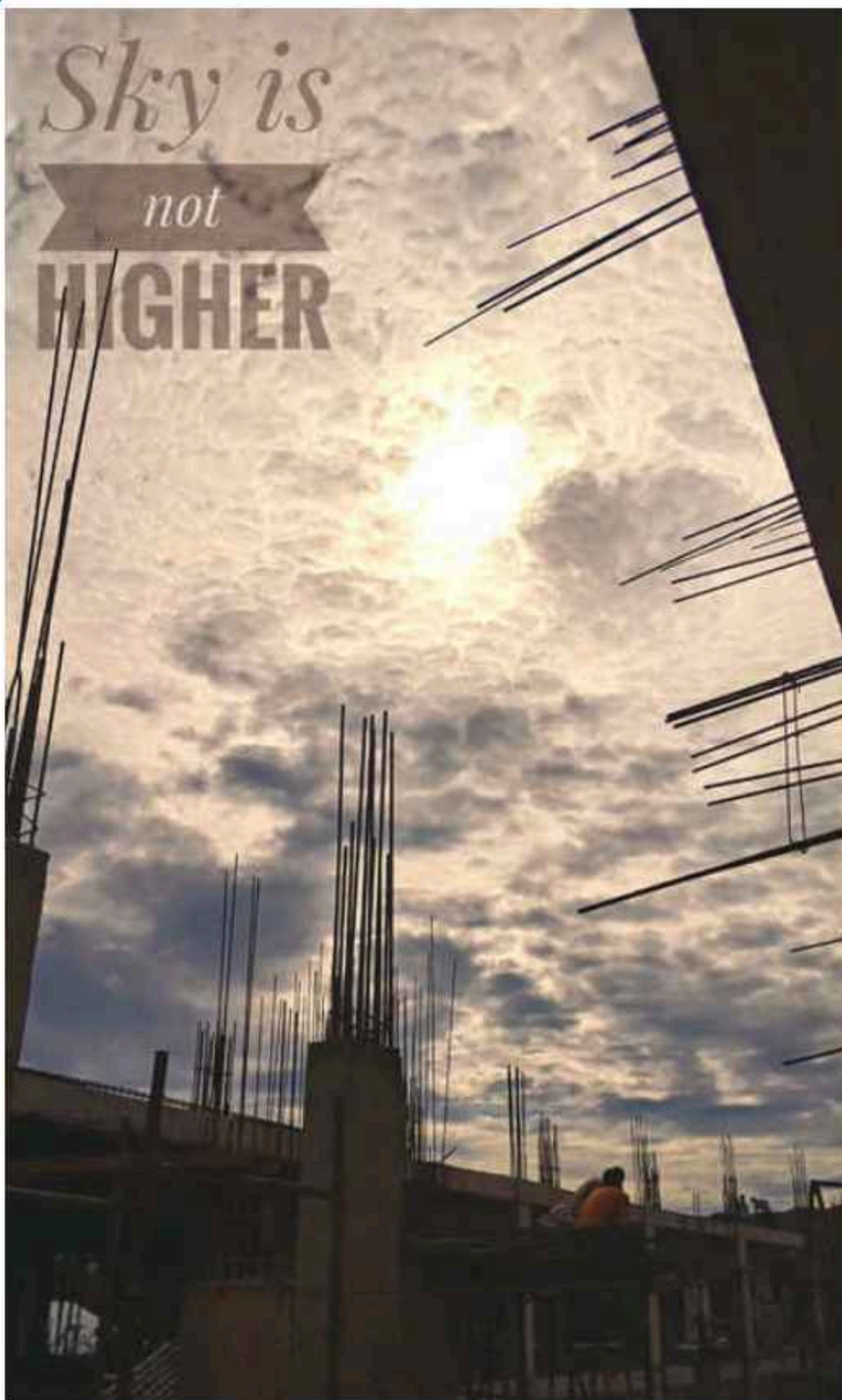




PHOTOGRAPH BY
VIJAY MALLIYA V S
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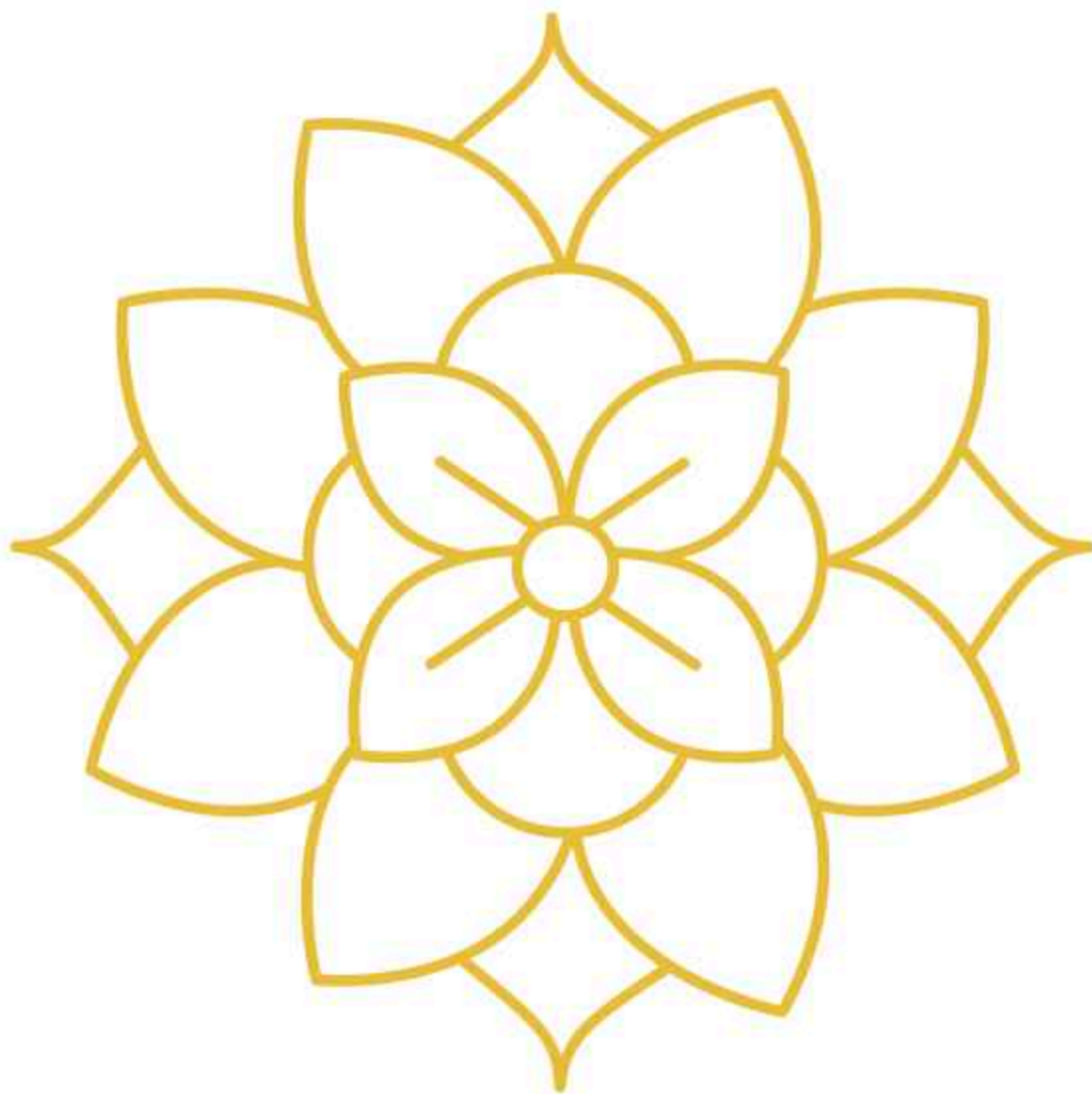


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R.Dhanusri/ I YEAR



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